NIGHT REALM

Game Design Document

2D Top-Down Hub-and-Dungeon RPG • Portfolio Edition

Designer	Trinh Quang Anh
Engine	Unity 2D — URP, DOTween, A* Pathfinding
Platform	PC (itch.io) — Mobile port proposed
Status	Shipped — playable on itch.io
Play	qanht.itch.io/night-realm
GitHub	github.com/T-qa/NightRealm

1. Executive Overview

Elevator Pitch

> ***You are an adventurer returning to a town on the edge of a dangerous dungeon. You rest, shop, gear up — then descend. The dungeon escalates, but so does your build. Every piece of equipment, every rune you slot, every level you gain makes you measurably stronger. The question is never "can I beat it?" The question is "how good can I make my build before the boss?"***

Night Realm is a 2D top-down hub-and-dungeon RPG where the player alternates between a safe town hub — shopping, equipment management, ability loadout — and an escalating dungeon of wave encounters, environmental hazards, and loot. Character progression is stat-driven, gear-dependent, and ability-shaped, giving players the satisfaction of a build coming together before the final boss confrontation.

Main Menu — New Game routes to Registration. Continue routes to Town. Clean atmospheric entry point.

MDA Framework

Layer	In Night Realm	Designer Implication
Mechanics	10-stat combat model, gold economy, rune equip system, equipment rarity rolls, wave-based spawning	Designer controls these directly. Every tuning decision lives here.
Dynamics	Build identity forming through gear + rune choices, risk/reward of spending gold vs saving for revive, level-up timing creating free heals mid-combat	Emerge from player behaviour. Cannot be scripted, only shaped by the mechanics.

Layer	In Night Realm	Designer Implication
Aesthetics	Hero growing stronger with every run. Satisfaction of a build validated by a boss clear.	The emotional target. All mechanics must serve this feeling.

Player Types & Motivations

Bartle Type	How Night Realm Serves Them	Primary Hook
Achiever	Clear the boss. Equip all Legendary items. Reach max level.	Visible stat numbers growing with every level-up. Rarity label on each item confirms collection progress.
Explorer	Equipment sub-effects are randomised — every shop visit offers something unseen. Rune combinations create emergent playstyle interactions.	What does this rune combination do? Drives experimentation across builds.
Competitor	Gold economy creates real spending decisions. Players who optimise their build through the economy outperform careless spenders.	Speed of boss clear and survival without revive as informal self-benchmarking.
Socializer	Not served in current build. No multiplayer.	Proposed: community boss challenge events for the mobile version.

2. Design Pillars

PILLAR 1 — EVERY STAT POINT MATTERS

The 10-stat combat model means no stat is irrelevant. Critical Chance, Block Chance, Accuracy, Evasion — they interact. A build with high Accuracy does not just hit more often: it also reduces the enemy's effective Block and Evasion against the player. Players who understand the interactions are measurably more effective. The game rewards engagement with its systems.

PILLAR 2 — THE HUB IS PART OF THE GAME

Town is not a loading screen. It is a strategic space. The shop's randomised stock creates purchasing decisions with real opportunity cost. The Town Chest creates cross-session inventory management. The death/revive decision ties the economy to risk tolerance. Every moment in Town is a design decision, not downtime.

PILLAR 3 — BUILD IDENTITY THROUGH CHOICE

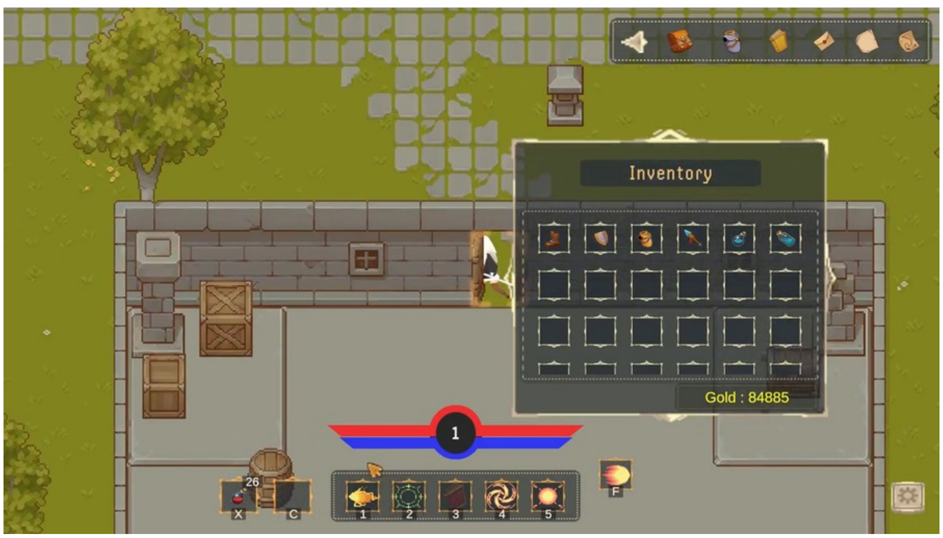
Five active ability slots + four passive slots + four gear slots + randomised sub-effects = distinct playstyle identities without a prescribed skill tree. A player who slots a healing rune and equips high-Block armor is playing a survival build. One who stacks Crit runes and Crit gear is playing an aggro build. Neither is told to do this — they discovered it.

3. Core Loop & Game Flow

Layered Core Loop

Loop Level	Cycle	Duration	Reward
Micro — per hit	Attack rolls: Crit → Block → Miss → apply damage	< 1 second	Hit/miss visual feedback, floating numbers, HP change
Meso — per stage	Enter zone → waves spawn → clear all enemies → chest unlocks	2–5 min	Gear, gold, and XP accumulation
Macro — per session	Town setup → Dungeon → Stages → Boss → Town return	20–40 min	Level-ups, build development, boss clear satisfaction

📊 **Full Game Flow — from Main Menu to boss defeated and return to Town** → Open in Whimsical



Town Hub — The safe strategic space between dungeon runs. Shopkeeper (top), portal to dungeon (bottom center).

Scene Structure

Scene	Contents	Design Role
Main Menu	New Game, Continue, Exit. Registration for new players.	Entry point. Continue only shows if a save file exists — prevents confusion.
Cutscene	Linear intro sequence. Plays once per new account only.	World-building and emotional context. Never repeated — respects returning players' time.
Town (Hub)	Shopkeeper, Town Chest, NPC dialogue, Portal to Dungeon.	Safe strategic space. All spending, loadout, and inventory decisions happen here.

Scene	Contents	Design Role
Dungeon Floor 1	Ambient monsters, Stage Encounter zones, Loot Chests, Spike Traps, Portal to Boss Map.	Risk space. Combat, loot, and hazards. Threat increases as player explores deeper.
Boss Map	Stone Golem boss (auto-heals to full on detection before Phase 1). One reward chest post-fight.	Climactic test of the player's build. Highest-stakes moment of every session.

4. Player Progression

Leveling System

XP is earned by killing enemies in encounter zones. The XP threshold grows exponentially — each level requires 1.5× more XP than the previous one. Early levels are fast (satisfying momentum), later levels are slower but more meaningful. Every level-up fully restores HP and Mana — a free heal that rewards sustained combat performance and creates a natural incentive to keep fighting rather than retreating.

📊 **Progression Loop — XP to Level Up to Stat Growth cycle** → Open in Whimsical

📊 **XP Scaling Curve — exponential threshold growth, ×1.5 per level** → Open in Whimsical

📊 **Stat Growth by Level — HP / Mana / Attack / Defence at L1, L3, L5, L7** → Open in Whimsical

Stat	Base at L1	Bonus Per Level	Value at L5	Value at L7
Max Health	3,000	+200	3,800	4,200
Max Mana	200	+100	600	800
Attack Power	100	+25	200	250
Defence	10	+10	50	70

Death & Revive

📊 **Death and Revive Decision — compact binary flow** → Open in Whimsical

Event	Outcome
Player HP reaches 0	Input disabled. Death animation plays. 2-second delay before prompt appears.
Revive prompt shown	"Pay 1,000G to continue here?" — player sees current gold balance.
Player pays 1,000G	Revives in same map. HP and Mana fully restored. Lose 1 level (minimum L1). XP resets to 80% of current threshold.
Player declines or cannot afford	Returns to Town for free. Same level and XP penalty applies regardless.

1,000G Revive — Economy Design in Action

1,000G is two-thirds of the starting gold (1,500G). A new player losing their first death spends most of their budget on a second chance. A mid-game player with 3,000G can afford three revives. A careless spender may not afford one. The cost is fixed — the player's relationship to it changes as the session progresses. This is economy as tension, not as barrier.

5. Combat System

The 10-Stat Model

Every combatant in Night Realm uses the same 10-stat model. Once a player learns that Accuracy reduces an enemy's effective Block and Evasion against them, that knowledge applies to every encounter. The system is fully consistent and fully learnable — it rewards investment in understanding it.

Stat	Controls	Design Note
Max Health	Total HP pool before death	Scales linearly with level. The most visible survivability stat.
Max Mana	Resource pool for active ability casts	Limits ability spam. Potions restore mana between stages.
Move Speed	World-unit movement per second	Fixed per entity type in current build — not a player-tunable stat.
Attack Power	Base damage multiplier for all abilities	All ability damage scales off this. Primary offensive stat.
Defence	Flat damage reduction on every incoming hit	Consistent reduction against all damage types. Often undervalued.
Block Chance	% chance to reduce an incoming hit by 70%	Effective Block = Block% minus attacker's Accuracy. Subject to Accuracy counterplay.
Accuracy	Reduces target's effective Block and Evasion when attacking	Offensive stat with defensive value — penalises enemies who try to block the player.
Evasion	% chance to dodge an incoming hit entirely	Highest ceiling defence. Effective Evasion = Evasion % minus attacker's Accuracy.
Critical Chance	% chance a hit deals bonus critical damage	Pairs directly with Critical Hit Damage. Neither stat is useful without the other.
Critical Hit Damage	Bonus multiplier applied on a critical hit	At 170%: critical hit deals 1.0 + 1.70 = 2.70× base damage.

Hit Resolution — 3 Sequential Checks

📊 **Combat Hit Resolution — Crit → Block → Miss, left to right** [→ Open in Whimsical]

1. Critical Check: if attacker's Critical Chance % roll succeeds → damage multiplier += (CritDMG ÷ 100). At 170% critical damage, multiplier becomes 2.70.
2. Block Check: if (Target Block Chance minus Attacker Accuracy)% roll succeeds → damage multiplier is reduced by 70%. Block still allows significant damage on a blocked critical hit.

3. Miss Check: if (Target Evasion minus Attacker Accuracy)% roll succeeds → effectively zero damage. Attacker's Accuracy directly reduces this chance.

Final damage = Raw Damage × final multiplier, clamped to a minimum of zero. Environmental damage from spike traps bypasses all three checks and always deals its exact configured value.

WORKED EXAMPLE — Blocked Critical Hit

Player: 100 Attack Power, 30% Critical Chance, 170% Critical Hit Damage. Enemy: 20% Block Chance. Player has 15 Accuracy, so enemy effective Block vs player = 20 minus 15 = 5%.

Step 1 — Critical: roll succeeds (30%). Multiplier becomes 1.0 + 1.70 = 2.70.
Step 2 — Block: roll succeeds (5% effective). Block removes 70% of damage: 2.70 minus 0.70 = 2.00.
Step 3 — Miss: roll fails. Hit connects.

Result: 100 × 2.00 = 200 damage. A blocked critical still deals twice base damage. Both Crit Chance and Block Chance were meaningful simultaneously — this interaction is intentional.

Combat — Slash ability against a Mushroom enemy. HP bar visible above target. Floating damage numbers confirm the hit.

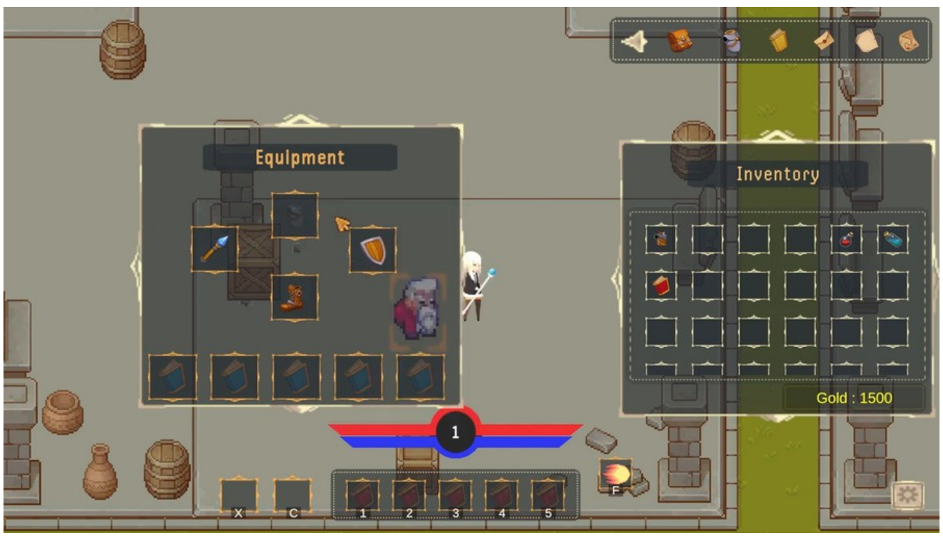
Combat — Fireball in flight. Full 5-slot skill bar with cooldown overlays visible at bottom of screen.

6. Ability System — Runes

How Runes Work

Runes are physical items that occupy ability slots — not passive bonuses from a skill tree, but actual items the player equips, swaps, and purchases. Build construction is tangible and reversible: unequip a rune at any time and its effect disappears immediately. The rune system is the primary driver of build identity.

Rune Type	Slots Available	Behavior	Resource Cost
Active	5 dedicated slots	Player activates with a key press. Has cast delay, cooldown, and optional mana cost. Deals damage, heals, or creates effects.	Mana (if any) consumed on cast + cooldown timer before reuse
Passive	4 dedicated slots (plus equipment bonus slots)	Always active while equipped. Permanently applies a stat buff or special effect.	None — always on while in slot



Equipment Panel — 4 gear slots at top (Weapon, Armor, Shoe, Shield). Passive rune slots below. Stats panel right side.

Ability Type	Cast Delay	Cooldown	Effect	VFX Color
Fireball (Fire damage)	Short (~0.5s)	Medium	Projectile dealing fire damage to first enemy hit	Red impact burst
Slash (Physical)	Minimal	Short	Melee-range burst attack	Yellow flash
Staff Heal (Self restore)	Medium (~1.0s)	Long	Restores player HP. Cannot overheal beyond max HP.	Green glow on player

Ability Type	Cast Delay	Cooldown	Effect	VFX Color
Storm / Explosion (AoE)	Long	Long	Area damage zone that persists briefly	Blue or orange AoE ring
Laser (Single-target)	Short	Medium	High single-target damage beam	Blue trail VFX

Known UX Gaps (v1.1 Backlog)

- "All 5 active slots full" — no feedback when player tries to equip a 6th active rune. Current build: silently does nothing. Fix: show "Slots full" floating text.
- "Not enough mana" — ability silently fails when mana is insufficient. Fix: show "Not enough mana" floating text immediately.
- "Inventory full during rune swap" — equip is blocked but player receives no explanation. Fix: show "Inventory full" popup before blocking the equip.

7. Equipment & Items

Equipment Slots

Four dedicated equipment slots: Weapon, Armor, Shoe, Shield. Equipping a new item automatically returns the previous item to inventory. No item is ever lost through the equip action — the swap is always safe and reversible.

Rarity System

📊 **Equipment Rarity Tier System — sub-effect count determines rarity** → Open in Whimsical

Sub-effects Rolled	Rarity	Border Color	Power Ceiling
0	Common	Grey	Base stats only. Reliable but no bonus potential.
1	Uncommon	Green	One bonus stat — slight power advantage over Common.
2	Rare	Blue	Two bonuses — meaningful advantage, build-enabling combinations possible.
3	Epic	Purple	Three bonuses — strongly affects playstyle direction.
4+	Legendary	Orange	Maximum sub-effects — same item name, very different power ceiling.

Equipment Tooltip — Uncommon armor shown with 1 sub-effect (passive Critical Chance bonus). Price visible for selling.

Stat Stacking Order

4. Flat additions applied first — e.g. +200 Attack Power from weapon main stat
5. Percentage additions applied second — e.g. +10% Critical Chance from sub-effect
6. Percentage multiplications applied last — e.g. ×1.15 from a passive rune

Order matters: multiplications applied last produce higher absolute gains when the base is large. Stacking flat bonuses first, then multiplying, is always more powerful than the reverse order. Players who understand this optimise more effectively.

Consumable	Effect	Stack Limit
Heal Potion (Light)	Restores a small percentage of Max HP	64
Heal Potion (Medium)	Restores a larger percentage of Max HP	64
Mana Potion (Light)	Restores a small amount of Mana	64
Mana Potion (Medium)	Restores a larger amount of Mana	64

8. Gold Economy

Economy Philosophy

Night Realm uses a single-currency economy: Gold. No premium currency, no secondary resource. Every economic decision is made in the same currency, making all trade-offs visible and legible. The revive system is the economy's most interesting design moment — it forces a real-time risk assessment every time the player dies.

📊 **Gold Economy — Sources feeding the wallet; Sinks drawing from it** → Open in Whimsical

Reward Schedule — Variable Ratio

Night Realm uses variable ratio reward scheduling throughout: equipment sub-effects are randomly rolled (0–4 per item), shop stock is randomised within item pools each visit, chest gold rewards vary within per-chest ranges. Variable ratio — where the player cannot predict exactly what they will receive — is the most psychologically engaging reward pattern. It drives the "one more stage" impulse that extends session length and retention.

System	Reward Type	Schedule	Player Psychology
Equipment sub-effects	Rarity tier (Common to Legendary)	Variable — 0–4 sub-effects randomly rolled	Every item could be Legendary. Near-miss on 3 sub-effects drives another shop visit.
Shop stock	Which items are available	Variable — randomised subset of pool each Town visit	Creates urgency: buy now or miss it. Stock never holds over between visits.
Chest gold reward	Gold amount	Variable — random within per-chest range	Consistent enough to feel reliable; variable enough to feel like a discovery.
Level-up stats	Stat growth amount	Fixed — same amount every level	Predictable progress signal. Anchors the variable rewards around it.

Gold Source	When Received
Starting Gold: 1,500G	New game only — once per save file
Loot Chest Gold Reward	First time each chest is opened in a dungeon run
Item Sales	Any time selling an item to the Shop NPC in Town

Gold Sink	Cost	Design Intent
Shop Purchases — Runes, Equipment, Consumables	Variable per item (100G–	Main economy drain. Drives the "save for something better or buy now?" tension.

Gold Sink	Cost	Design Intent
	600G typical range)	
Revive Fee	1,000G fixed	High-stakes binary decision. Fixed cost with variable emotional weight depending on current gold balance.

9. Enemy Design & Roster

Goblin — Melee, fast

Mushroom — Melee, AoE

Flying Eye — Ranged

Enemy	Archetype	Behavior	Threat	Counter-play
Goblin	Melee — Fast	Charges directly at player using A* pathfinding. Fast movement.	HP damage on contact. Speed makes maintaining distance difficult.	Kite. Prioritise over slower enemies. Fireball works well at range.
Mushroom	Melee — Medium	Similar movement to Goblin but slower. AoE zone ability.	Poison zone punishes stationary players. Forces movement.	Keep moving at all times. Never stop during a Mushroom stage.
Flying Eye	Ranged	Maintains distance from player. Fires projectiles. Reused as boss minion.	Projectile damage forces player to advance into boss range.	Close the gap aggressively. Range abilities are effective.

Enemy Type	Respawn?	Player Choice
Ambient Monsters — roam dungeon freely, attack if player enters range	Yes — continuous up to max count	Optional combat. Useful for XP grinding between stages.
Stage Encounter Enemies — spawn in waves when player enters trigger zone	No — cleared stages stay cleared	Required for chest unlock. Camera locks — cannot retreat once triggered.
Boss Minions — summoned by Stone Golem in Phase 2	No — die permanently	Must clear to reduce incoming damage pressure. Reuse Flying Eye sprite.

10. Stage Encounters & Boss Fight

Stage Encounter Flow

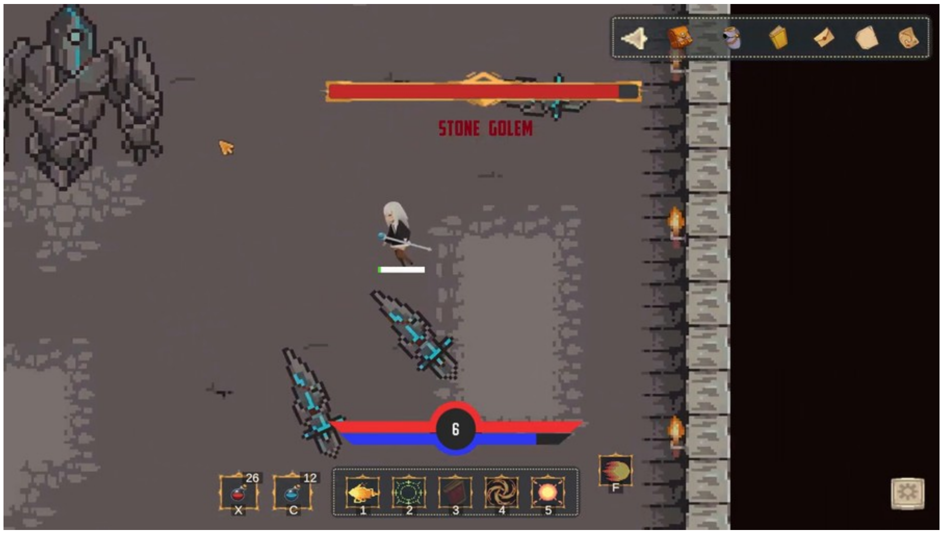
7. Player steps into Stage Trigger zone — floor-level trigger, invisible to player
8. Camera locks to the arena — player cannot leave the stage area
9. Wave 1 spawns at designated spawn points around the arena
10. All Wave 1 enemies cleared → brief pause → Wave 2 spawns
11. All waves cleared → Stage Clear banner → camera unlocks → adjacent chest activates

"STAGE CLEAR" — camera unlocks immediately after this display. Adjacent loot chest becomes collectible.

Boss Fight — Stone Golem

The Stone Golem auto-detects the player on entry and heals to full HP before Phase 1. Any damage dealt before detection is cancelled — the boss always starts Phase 1 at 100%. This is intentional: the player should not be able to whittle the boss down during the approach animation.

Phase	HP Range	Abilities	Counter-play Strategy
Phase 1	100% to 50%	Targeted projectile attacks. Fixed patterns. Attack timing is consistent.	Learn the attack timing. Punish during the wind-up animations.
Phase 2	Below 50%	Laser Fire (rotating cyan beam sweeping the arena), Fire Orbs, Spawn Monster (Flying Eye adds each cycle).	Kill Flying Eye adds first. Move constantly to avoid the laser sweep. Burst boss during ability transitions.



Boss Fight — Phase 1. Stone Golem name confirmed in HP bar. Projectile attack in flight. Player must dodge and counterattack.

Boss Phase 2 — Laser Fire ability. Rotating cyan beam sweeps the full arena. Continuous movement required to survive.

11. Difficulty Curve

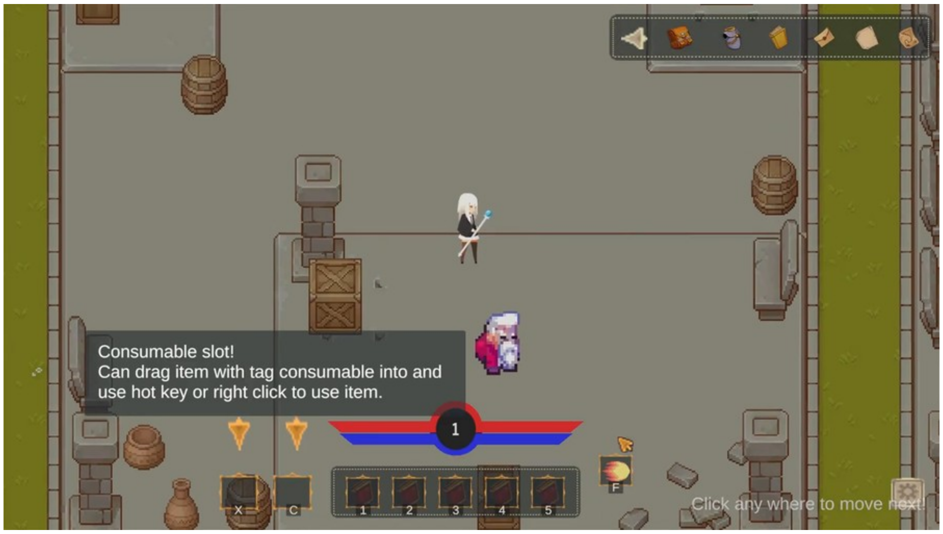
The session arc moves from Town setup — low pressure, strategic — through escalating dungeon stages to the boss peak. Boss Phase 2 is the hardest moment in the game, and it is earned by clearing everything before it. A player who built well and spent wisely will succeed. A player who ignored the economy will struggle.

📊 **Difficulty Curve — Town → Early Stages → Hard Stages → Boss P1 → Boss P2**
→ Open in Whimsical

Phase	Location	Difficulty	Primary Design Lever
Setup	Town	None — pure strategy	Player spending decisions, rune loadout, consumable stocking. No combat.
Early Stages	Dungeon	Low to Medium	Goblin + Mushroom. Player learns stage encounter flow. XP accumulates quickly.
Hard Stages	Dungeon	Hard	Flying Eye mix adds ranged pressure. Mana management for abilities becomes critical.
Boss Phase 1	Boss Map	High	Pattern recognition. Attack timing. The build assembled in Town is fully tested.
Boss Phase 2	Boss Map	Very Hard	Multitasking: clear adds vs burst boss. Economy decision (mana potions) made in Town now matters.

12. UI/UX Design

HUD Layout



FTUE Tutorial Overlay — consumable slot highlighted. Full HUD visible underneath. Skill bar visible with cooldown states at bottom.

HUD Zone	Element	What It Communicates
Top-left	HP Bar (red fill)	Current HP / Max HP. Depletes permanently until revive or heal item used.
Below HP bar	Mana Bar (blue fill)	Current Mana / Max Mana. Consumed by active ability casts. Potions restore.
Top-right	Level + Gold	Current player level and gold amount. Updates in real time.
Bottom-center	Skill Bar	5 active ability slots + 2 consumable quick-use slots. Cooldown overlay on each active slot.
Full-screen (boss)	Boss Health Bar	Large HP bar during boss fight. Boss name shown above. Highest visual priority in the game.
Floating	Combat Numbers	Damage, Heal, MISS, and BLOCK text floating above entities. Immediate hit resolution feedback.
Center (stage)	Stage Indicator	Stage start and "STAGE CLEAR" banners. Shows current wave within stage.

Shop and Inventory open simultaneously. Items shown with prices. Player gold displayed in inventory panel.

FTUE Design Rules

FTUE Design Principles for Night Realm

Rule 1: Never pause gameplay to show a tutorial popup. All overlays appear while the game is running — informational, not gates.

Rule 2: Each FTUE tooltip covers exactly one mechanic. Never combine two lessons in one overlay.

Rule 3: Tooltips triggered by real moments (first shop visit, first level-up, first stage encounter) have far higher retention than front-loaded text.

Rule 4: Every tooltip auto-dismisses after 5 seconds and never reappears — the game respects the player's intelligence and assumes the lesson was absorbed.

13. Feature Spec — Rune Equip System

Feature Specs define a single system for Designer + Developer + Artist collaboration.

Field	Detail
Feature Name	Rune Equip System
Purpose	Allow players to build distinct ability identities by equipping rune items into dedicated slots. The intersection of rune loadout and equipment sub-effects creates playstyle variety without a prescribed skill tree.
Player Experience	Agency and ownership. The rune build should feel like the player's creation — not a path handed to them. Swapping runes between runs is a strategic ritual, not a chore.
Slots	5 Active Rune slots (key-press abilities with cooldowns) + 4 Passive Rune slots (permanent buffs while equipped). Equipment can add bonus passive slots via sub-effects.
Equip Logic	If target slot is empty: rune moves directly into slot. If slot is occupied: swap — equipped rune returns to inventory, new rune takes the slot. If inventory is full during a swap: equip is blocked (not enough room for the displaced rune).
Cooldown on First Equip	Active rune equipped for the first time starts on its standard cooldown. Prevents the equip-then-immediately-use exploit.
Edge Cases	Slot full when equipping a 6th active rune: silently blocked. UX gap — add "Slots full" floating text in v1.1. // Player dies mid-cast: cast cancelled, ability does not fire, cooldown does NOT trigger. // Not enough mana: silent fail. UX gap — add floating text in v1.1. // Passive rune unequipped: stat buffs removed immediately on unequip.
Open Question	Should there be a "rune preview" mode — equip a rune, see stat changes before confirming? Low engineering cost, high clarity value for build decision-making during Town phase.

14. Audio & Visual Direction

Context	Visual Style	Lighting
Town	Warm, pastoral, safe. 6-layer parallax hills in the background.	Bright and open. Highest contrast ratio in the game — signals safety.
Dungeon	Dark stone tileset. Environmental hazards (spike traps, cyan pool).	Rooms dark on first entry, fade to lit over 0.5 seconds on visit.
Boss Room	Tense, minimal. Dungeon tileset but cleared of clutter.	Fully lit on entry. Boss health bar takes visual priority over all HUD.

Dungeon Room Layout — Stone floor tiles, stage encounter area with side alcoves. Cyan hazard pool visible.

Color	Damage Type	Examples
Red	Fire	Fireball impact explosion, fire-type enemy attacks
Blue	Magic / Arcane	Flying Eye projectile hit, Laser ability trail and impact
Green	Poison / Nature	Mushroom AoE ground zone, nature-type status effects
Yellow	Physical / Critical Hit	Critical hit spark, general physical damage feedback
Purple	Dark / Arcane Variant	Specific spell hit effects, dark-type attack impacts

15. Mobile Adaptation & LiveOps

Mobile Monetization Strategy

📊 **Mobile Monetization Funnel + Revenue Model — Hub-and-Dungeon RPG F2P** → Open in Whimsical

Revenue Stream	Model	Estimated Revenue Share	Hard Design Rule
Equipment Gacha-Lite	Pull for random equipment. Guaranteed Rare every 10 pulls.	~35%	Gacha pool cannot contain stats unavailable to F2P players. Same max stat ceiling for all.
Battle Pass (Season)	Monthly cosmetic + gold reward track. Free tier always available.	~30%	Free tier gives meaningful gold and potions — not just decorative cosmetics.
Gold IAP	Convenience acceleration only. Soft cap: buy up to daily F2P earn amount.	~20%	Cannot create a gold gap large enough to purchase advantages unavailable to free players.
Cosmetic IAP	Character outfits, town decorations, rune visual effects.	~10%	Visual only — zero stat difference from base versions.
Energy System	10 free dungeon runs per day. Full restore in 6 hours, or 1 paid refresh per day max.	~5%	Core gameplay is never fully blocked. Free players can always play.

HARD DESIGN RULE — Build Integrity

Equipment from gacha cannot have exclusive stats unavailable to F2P players. A paying player and a free player of the same level must have the same maximum stat ceiling. Money accelerates progression — it does not grant an exclusive power floor that free players cannot reach through play.

KPI	Target	Action If Below Target
D1 Retention	40%+	Review FTUE. D1 drop-off is almost always confusion or boredom in the first session.
D7 Retention	20%+	Review daily challenge and Battle Pass week-1 rewards. These should drive days 3–7 habit.
D30 Retention	8%+	Review LiveOps event cadence. Need one meaningful event every two weeks minimum.
Session Length	7–12 min	Too short: loop not engaging. Too long: fatigue — shorten dungeon floor depth.

16. Balancing Values Reference

Player Base Stats (Level 1)

Stat	Value	Notes
Max Health	3,000	+200 per level
Max Mana	200	+100 per level
Move Speed	3.0	Fixed — not player-tunable in current build
Attack Power	100	+25 per level
Defence	10	+10 per level
Block Chance	20%	Fixed base — modified by equipment sub-effects
Accuracy	15	Fixed base — modified by equipment sub-effects
Evasion	20%	Fixed base — modified by equipment sub-effects
Critical Chance	30%	Fixed base — modified by equipment and runes
Critical Hit Damage	170%	2.70× damage on critical hit

Economy Values

Parameter	Value
Starting Gold	1,500 G
Revive Cost	1,000 G (fixed)
Gold Minimum	0 G — cannot go negative
Levels Lost on Death	1 level (minimum Level 1)
XP After Death	Reset to 80% of current level threshold
Town Chest Slots	9 slots (persistent across all sessions)
Inventory Slots	30 slots
Consumable Stack Limit	64 per stack

17. Risks & Open Questions

Risk	Likelihood	Impact	Mitigation
"Silent fail" on full slot and no mana — player does not know why ability did not fire	High	Medium	Add floating text for both conditions in v1.1. Low engineering cost, high UX value.
"Silent discard" when inventory auto-fills — player may feel items were stolen	Medium	High	Add "Inventory full" popup before any discard. Never silently destroy an item the player has not seen.
Single dungeon floor limits replayability — players who clear boss have no reason to return	High	High	Content roadmap: Floor 2 with new enemy types + boss. Limited-time event dungeons for mobile LiveOps.
Equipment gacha perceived as pay-to-win if pool design is not careful	Medium	High	Strict pool rule: zero exclusive stats in paid pool. Same max ceiling for all players — enforced at design stage, not just policy.

- Difficulty select: Easy/Normal/Hard changing enemy HP and damage scalars. Low engineering cost, high accessibility value for new players.
- Shop stock refresh timing: should stock refresh on each Town visit or require a manual action? Current behavior is unclear — needs explicit designer decision before v1.1.
- Cutscene skip: if runtime exceeds 90 seconds, add a skip option. New player patience for non-interactive content is limited.

18. UI/UX Wireframe Reference

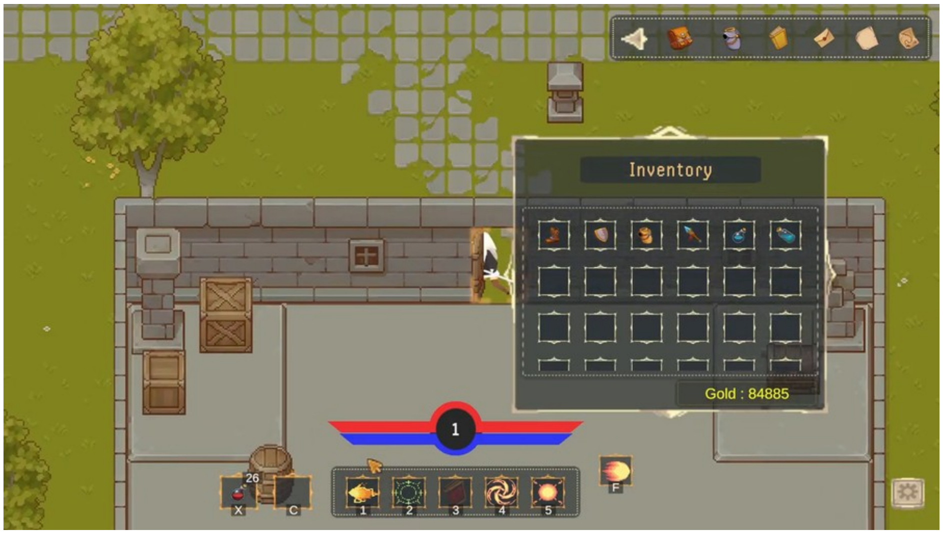
Wireframe Coverage

Night Realm's UI wireframes cover the 8 screens the player sees most frequently, designed to show layout logic, information hierarchy, and interaction decisions. All wireframes are built in Whimsical as annotated wireframe layouts.

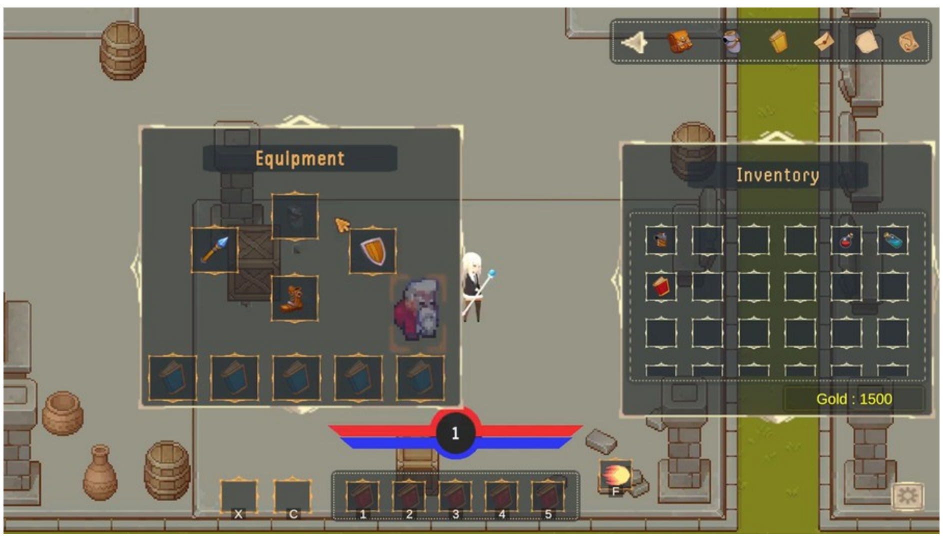
Screen	What It Shows	Key Design Decisions
01 — Main Menu	Game title, New Game, Continue, Quit buttons	Continue disabled if no save file exists — prevents confusion on first launch
02 — Registration	Name input modal, Start Adventure CTA	Name is used for save file label and leaderboard. Max 16 characters.
03 — Town Hub	Shop, Town Chest, and Portal positions. Level and Gold HUD strip.	All strategic decisions happen here. No combat in Town at any time.
04 — Dungeon HUD	HP/Mana bars, Level/Gold, 5-slot skill bar with cooldowns, stage indicator	Skill bar is bottom-center — reachable by both thumbs on mobile. Stage indicator shows wave progress.
05 — Equipment Panel + Stats	4 gear slots, 4 passive rune slots, 10-stat display with current values	Stats update in real time when gear or runes are swapped. Shows XP progress to next level.
06 — Boss Fight HUD	Full-width boss HP bar above skill bar. Phase 1/2 indicator.	Boss bar replaces stage indicator during boss fight — highest visual priority.
07 — Death & Revive Prompt	Gold check (enough vs broke), Revive CTA, Return to Town CTA	Broke state disables Revive CTA. Town return always available and always free.
08 — Boss Victory Popup	"Boss Defeated" confirmation, reward chest reference, two CTA buttons	Collect Reward vs Return to Town — chest persists if player skips and returns later.

Whimsical Wireframe Board: [NR — UI/UX Wireframes → Open in Whimsical]

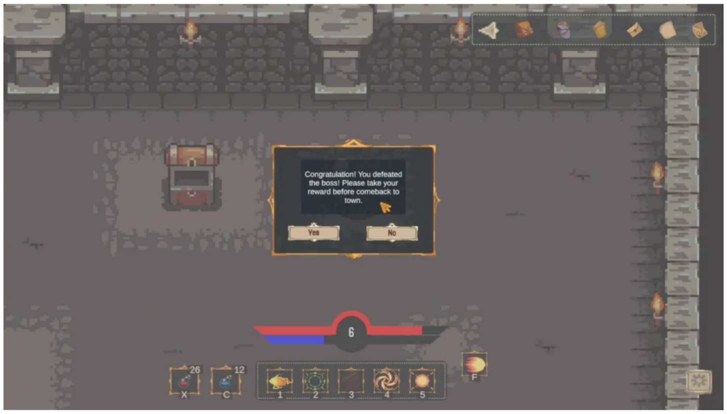
Key Screen Previews



Town Hub — Safe strategic space. Shopkeeper NPC (top), portal to dungeon (bottom center). All non-combat decisions happen here.



Equipment Panel — 4 gear slots (top) + passive rune slots (bottom-left). Character stat panel (right) shows all 10 stats.



"Congratulations! You defeated the boss!" — Build validated. Reward chest available before returning to Town.

19. Appendix — Diagram Index

Diagram	What It Shows	Whimsical Link
Full Game Flow	Complete scene flow from app launch to boss defeated and Town return	whimsical.com/nr-full-game-flow-EtWNDVCdnLre97hkYf4SN4
Combat Hit Resolution	3-step Crit → Block → Miss sequence with formula and worked example	whimsical.com/nr-combat-hit-resolution-SzH3S2W9JkXhLR3Nb4azVf
Death and Revive Decision	Compact binary flow: die → can afford 1000G? → revive or Town	whimsical.com/nr-death-and-revive-decision-Vh6GvvuSAhoRRqEwZPwkdA
Progression Loop	XP → Level Up → Stat Growth cycle with gear and rune inputs shown	whimsical.com/nr-progression-loop-CswPkEMkNgr54SCTXmEdDj
XP Scaling Curve	Exponential XP threshold growth, ×1.5 per level, shown as progression chain	whimsical.com/nr-xp-scaling-curve-1-5-per-level-7MjUbwwvxB98JCCZz9jz11
Stat Growth by Level	HP, Mana, Attack, Defence values at L1, L3, L5, and L7	whimsical.com/nr-stat-growth-by-level-l1-l3-l5-l7-ovsA59qzm1gQyMT3FiR4s
Equipment Rarity Tiers	Sub-effect count to rarity tier chain (Common to Legendary)	whimsical.com/nr-equipment-rarity-tier-system-7H2yfrWFEeBUEdSZqT3JBF
Gold Economy — Sources & Sinks	Sources flowing into wallet; sinks flowing out — with design annotation	whimsical.com/nr-gold-economy-sources-and-sinks-5Q1SMmCuBsAPAcAxoLMJ9Z
Difficulty Curve	Town (low) → Early Stages → Hard Stages → Boss P1 → Boss P2 (peak)	whimsical.com/nr-difficulty-curve-town-stages-boss-H2sk7btsnJNSPJ6h7AsN3K
Mobile Monetization Funnel	Player funnel (Install → D1 → D7 → Payer) + F2P revenue model breakdown	whimsical.com/nr-mobile-monetization-funnel-revenue-model-QEcKtfTp6sd9Wxg6vXUs7P
UI/UX Wireframes	8 key screens: Main Menu, Registration, Town, Dungeon HUD, Equipment, Boss, Death, Victory	whimsical.com/nr-ui-ux-wireframes-5ACW1z8eFFuLxX5Ha2ba9i

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